

Jiwoo Shin

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Research Interests

I am interested in **inference-time controllability and alignment** of generative models, focusing on methods that steer models toward safe and user-aligned outputs without requiring expensive retraining.

My recent research focuses on **safe generation in discrete diffusion models**, developing guidance methods that operate at inference time.

Education

Korea Advanced Institute of Science and Technology , Daejeon, South Korea M.S. student, Kim Jaechul Graduate School of AI	09/2026 – Present
• Advisor: Prof. Il-Chul Moon	
Korea Advanced Institute of Science and Technology , Daejeon, South Korea B.S. in Industrial and Systems Engineering & Computer Science (Double Major)	03/2019 – 08/2026
• 2 years leave of absence for mandatory military service (10/2022 – 07/2024) • GPA: 3.87/4.3, Major GPA: 4.03/4.3	
University of Portland (UP) , Portland, OR	08/2025 – 12/2025
• Exchange Student, Department of Computer Science • GPA: 4.0/4.0 (Dean's List)	
Chungnam Science High School , Gongju, South Korea	03/2017 – 02/2019
• Early Graduation	

Research Experience

KAIST AAILab (Advisor: Prof. Il-Chul Moon), Daejeon, South Korea	12/2021 – 08/2026
• 01/2026 – 08/2026: Inference-time control of discrete diffusion models • 03/2025 – 07/2025: Training-free safe image generation in text-to-image diffusion models • 09/2024 – 12/2024: Retrieval-augmented generation (RAG) • 12/2021 – 08/2022: Probabilistic deep generative models; knowledge distillation; class-imbalanced learning	
University of British Columbia (Advisor: Prof. Mijung Park), Remote	07/2026 – 08/2026
• Focus: Safe text generation in uniform-state diffusion models	
University of Chicago (Advisor: Prof. Jooyoung Sohn), Remote	10/2025 – 12/2025
• Focus: Improving treatment planning and delivery in HDR-IORT	

Industry Experience

Spidercore Inc. , Daejeon, South Korea	12/2024 – 02/2025
• AI Research Intern • Task: Predicting ADMET properties of molecules through GNN models pretrained in a multimodal setting	

Publications

[3] **Prompt-Based Safety Guidance Is Ineffective for Unlearned Text-to-Image Diffusion Models**

Jiwoo Shin, Byeonghu Na, Mina Kang, Wonhyeok Choi, Il-Chul Moon

NeurIPS 2025 Workshop on Generative and Protective AI for Content Creation.

[2] **Training-Free Safe Text Embedding Guidance for Text-to-Image Diffusion Models**

Byeonghu Na, Mina Kang, Jiseok Kwak, Minsang Park, **Jiwoo Shin**, SeJoon Jun, Gayoung Lee, Jin-Hwa Kim, Il-Chul Moon

Neural Information Processing Systems (NeurIPS), 2025.

[1] **Nodes-in-Cluster Significance Score (NCSS): Keyword Extraction with Redefined TW-IDF using Stock News Text Data**

Junhyeok Jung*, Kwanghyeon Lee*, **Jiwoo Shin***, Jieun Han*

Korea Computer Congress (KCC), 2022 (poster).

* denotes equal contribution.

Awards

Magna Cum Laude , KAIST	08/2026
Next-Generation Engineer (Grand Prize), IPESK	09/2025
Dean's List (Top 3%), KAIST ISE	Fall 2024, Fall 2020
Basic Military Training Honor Graduate (Ranked #3), ROKAF, Class 842	12/2022
IE Frontier Encouragement Award, KAIST ISE	11/2022

Scholarships

Korea-U.S. Advanced Technology Youth Exchange Program (9,000 USD), Minister of Trade, Industry and Energy	Fall 2025
Department Honors Scholarship (Ranked #2), KAIST ISE	Spring 2022
Department Honors Scholarship (Ranked #1), KAIST ISE	Fall 2020

Activities

KAIST Buddy Program , KAIST ISSS	Spring 2026, Fall 2024
• Assisted exchange students with campus life and cultural adaptation	
Student Council , KAIST ISE	2020 – 2021
• 2021: Vice President	
• 2020: Vice Representative	